AI-Powered CV Parser: Technical Report

1. Executive Summary

The **AI-Powered CV Parser** is a sophisticated tool designed to automate the extraction of structured information from unstructured resumes and curriculum vitae (CVs). By leveraging state-of-the-art Large Language Models (LLMs) and robust document processing pipelines, the system transforms complex PDF and image-based documents into standardized JSON format, significantly reducing the manual effort required in recruitment and HR workflows.

2. Core Objectives

The primary goal of this project is to provide a reliable, scalable, and accurate solution for:

- **Automated Data Extraction**: Identifying and extracting key candidate details such as contact information, education, and professional experience.
- **Structured Output Generation**: Ensuring that the extracted data adheres to a predefined schema for easy integration with Applicant Tracking Systems (ATS).
- **Multi-Format Compatibility**: Supporting both text-based PDFs and scanned image-based resumes through Optical Character Recognition (OCR) techniques.

3. Technical Architecture

3.1 Large Language Model (LLM)

The system utilizes the **Mistral-Nemo-Instruct-2407** model, a high-performance LLM optimized for instruction-following tasks. This model provides the “intelligence” required to understand the semantic context of various resume layouts and terminologies.

3.2 Frameworks and Libraries

The project is built upon a modern technology stack:

Component	Technology
Language Model Interface	Hugging Face Transformers
Orchestration	LangChain
Web Interface	Streamlit
Deep Learning Backend	PyTorch
Document Parsing	PyPDF, OCR Utilities

3.3 Data Schema

The parser is configured to extract the following structured fields:

- **Full Name**: The candidate's legal name.
- **Email**: Primary contact email address.
- **Education**: A list of academic qualifications, institutions, and dates.
- **Skills**: A comprehensive list of technical and soft skills.
- **Experience**: Detailed professional history, including roles, companies, and responsibilities.

4. System Workflow

1. **Document Ingestion**: The user uploads a CV (PDF or Image) via the Streamlit dashboard.
2. **Preprocessing**: The system extracts raw text from the document. For images, OCR is applied to convert visual text into machine-readable strings.
3. **Prompt Engineering**: A structured prompt is generated, incorporating the raw text and specific formatting instructions for the LLM.
4. **LLM Inference**: The Mistral-Nemo model processes the prompt and generates a response containing the extracted data.

5. **Parsing & Validation**: LangChain's `StructuredOutputParser` validates the LLM's output against the defined schema and ensures a clean JSON block is returned.
6. **Visualization**: The final structured data is displayed to the user in an interactive format.

5. Key Features

- **Intelligent Understanding**: Unlike regex-based parsers, this system understands context, allowing it to handle diverse resume formats.
- **Scalable Design**: The modular architecture allows for easy updates to the model or the extraction schema.
- **User-Friendly Interface**: The Streamlit-based UI provides a seamless experience for non-technical users.

6. Conclusion

The AI-Powered CV Parser represents a significant advancement in HR technology. By combining the power of LLMs with structured parsing techniques, it offers a robust solution for modernizing the recruitment process. Future enhancements may include support for more languages, deeper sentiment analysis of candidate descriptions, and direct integration with popular HR platforms.

Report generated for the AI CV Parser Project